The Handshake Problem

– Space Complexity (Detailed Analysis)

Program:

```
// Handshake Problem Recursion

// This code calculates the number of handshakes that occur
when a given number of people are in a room. It uses
recursion to calculate the number of handshakes based on the
number of people.
#include <iostream>
using namespace std;

int handshake(int n) // Function to calculate the number of
handshakes
{
    if (n == 0 || n == 1) // Base case
    {
        return 0; // If there is only one person in the
room, there are no handshakes
    }
    cout << n << endl;
    return handshake(n - 1) + (n - 1); // Recursive call :
n-1 people shake hands with each other and the remaining
person shakes hands with n-1 people
}

int main() // Main function
{
    int n;
 // Number of people
    cout << "Enter the number of people: ";
  // Input the number of people
    cin >> n;
  // Store the number of people
    cout << "The number of handshakes are: " <<
handshake(n); // Output the number of handshakes
```

```
    return 0;
  // Return statement
}
```

$$\textit{Space Complexity}$$

$$A.\textit{Input Space Complexity}$$

1. Parametric Input Space

- $\textit{Only parameter is } n \ (\textit{an integer})$
- $\textit{Size of an integer} = \textit{constant} \ (e.g., 4 \textit{ bytes})$
- $\textit{No extra input structures} \ (\textit{arrays, lists, etc.})$

$$\textit{Parametric Input Space} = \Theta(1)$$

2. Full (Holistic) Input Space

$\textit{Now think deeper}$:

$$\textit{What is the real} - \textit{world input of the handshake problem}?$$

- $\textit{The problem says: } n \textit{ people in a room}$
- $\textit{If we explicitly represent them, we might store}$:

```
Person people[n];
```

$\textit{That would take}: O(n)$

$But\ according\ to\ the\ program\ ,it\ only\ have$:

```
    int n;
```

So:

- $Logical\ problem\ input\ (Holistic\ Input) \rightarrow\ \Theta(n)$
- $Program\ input\ representation(Parametric\ Input)\ \rightarrow\ \Theta(1)$

$Therefore\ ,based\ on\ program - level\ i.e.according\ to\ the\ code\ we\ will\ take,$

$Parametric\ Input\ as\ `final\ Input\ Space\ `\ i.e.\Theta(1).$

$Auxiliary\ Space$

$Includes$:

- $recursion\ stack$
- $local\ variables$
- $temporary\ data$

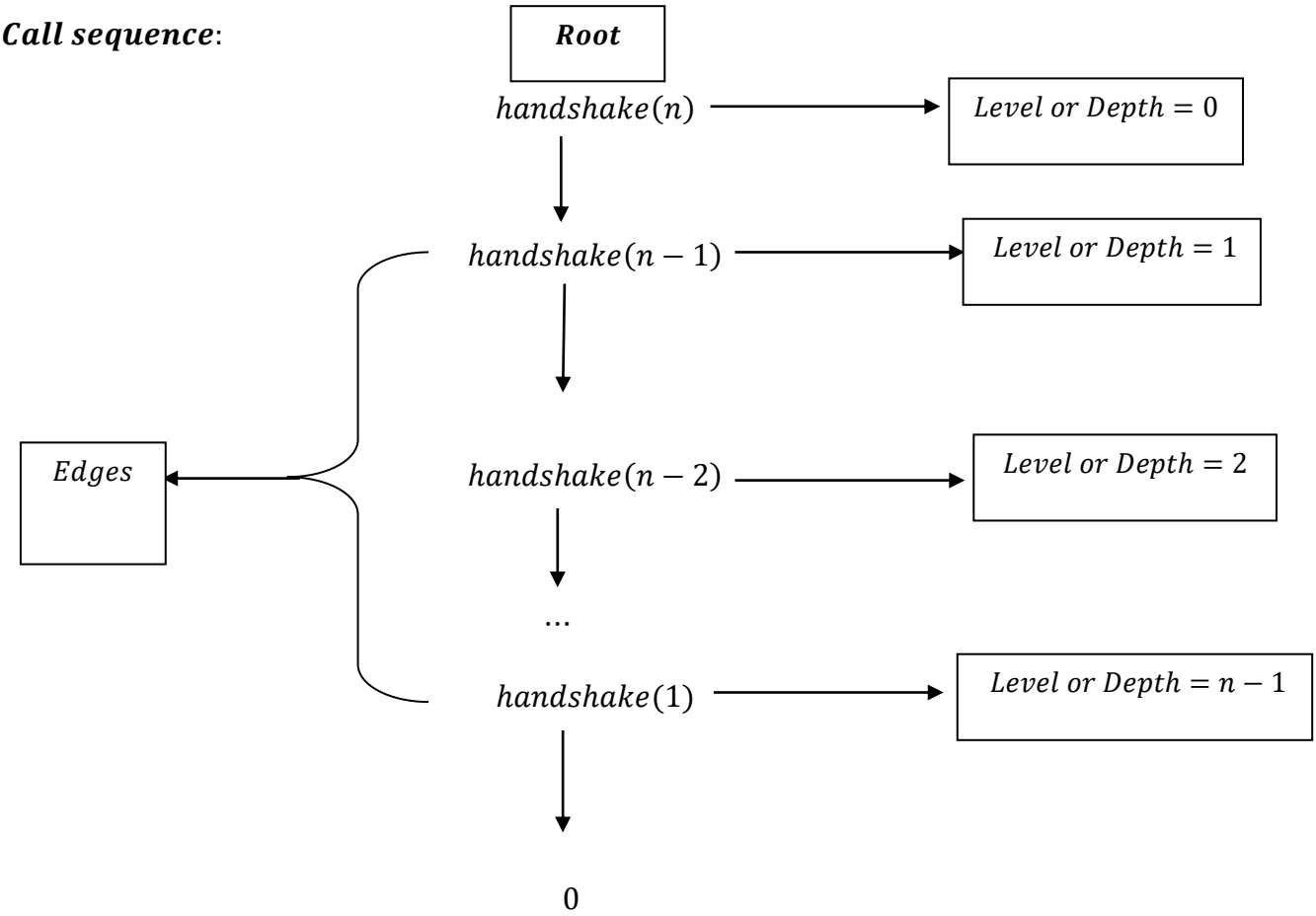
$Each\ recursive\ call\ creates\ a\ stack\ frame\ containing$:

- $parameter\ n$
- $return\ address$
- $temporary\ value\ (n-1)$
- $I/O\ context\ for\ cout$

$Each\ frame\ =\ \Theta(1) space\ per\ recursive\ call$

Recursion Structure

Call sequence:



$$\boldsymbol{Nodes(N) = Edges(k) + 1(Root)}$$

$$\boldsymbol{= (n-1) + 1 = n}$$

$$\boldsymbol{n - k(Edges) = 1\ [H(1), Base\ Case\ ] \Longrightarrow k = n - 1\ edges.}$$

$$\boldsymbol{Here\ n \rightarrow represents\ user\ input`n`\ i.e.\ n\ of\ H(n).}$$

$$\boldsymbol{Hence\ , we\ take\ here\ the\ Maximum\ Depth = no.of\ Edges}$$

$$\boldsymbol{= n - 1}$$

$\boldsymbol{Based\ on\ Active\ stackframe}$:

$$\therefore \boldsymbol{Total\ Auxiliary\ Space}$$
$$= (\boldsymbol{Maximum\ Depth} + \mathbf{1}) \times \begin{pmatrix} \boldsymbol{Space\ per} \\ \boldsymbol{Recursive\ Call} \end{pmatrix}$$

$$\begin{bmatrix} \boldsymbol{Here + 1\ for\ Root\ Node\ , which\ is\ also\ an\ active} \\ \boldsymbol{stack\ frame} \end{bmatrix}$$

$$= \boldsymbol{\Theta}(\mathbf{n} - \mathbf{1} + \mathbf{1}) \times \boldsymbol{\Theta}(\mathbf{1})$$
$$= \boldsymbol{\Theta}(\mathbf{n}) \times \boldsymbol{\Theta}(\mathbf{1})$$

$$= \boldsymbol{\Theta}(\boldsymbol{n})$$

$\boldsymbol{Total\ Space\ Complexity}$:

$$\boldsymbol{Input\ Space + Auxiliary\ Space} = \boldsymbol{\Theta}(\mathbf{1}) + \boldsymbol{\Theta}(\boldsymbol{n})$$
$$= \boldsymbol{\Theta}(\boldsymbol{n}).$$